

12. Use Figma to create an interactive diagram illustrating the evolution from 1G to 5G.

Design: -





Evolution Of 1G

1. Introduction of 1G:
 - The first commercially available 1G network was launched in Japan in 1979 by NTT (Nippon Telegraph and Telephone).
 - This generation of mobile networks introduced analog voice communication and primarily provided mobile voice services.
2. AMPS Standard (Advanced Mobile Phone System):
 - In the United States, the Advanced Mobile Phone System (AMPS) became the prevalent 1G standard.
 - AMPS used analog modulation for voice transmission and provided basic voice calling services.
3. Limited Data Services:
 - 1G networks were primarily designed for voice communication, and data services were extremely limited.
 - Data services were characterized by low data transfer rates, typically used for simple messaging services.
4. First Mobile Phones:
 - The introduction of 1G saw the emergence of the first generation of mobile phones. These devices were bulky and had limited battery life compared to modern standards.
5. Global Expansion:
 - Following the initial deployment in Japan and the United States, 1G networks expanded globally, reaching other countries in Europe and Asia.
6. Challenges:
 - 1G networks faced challenges such as limited capacity, poor call quality, and susceptibility to interference and eavesdropping.
7. Transition to Digital Technologies:
 - Towards the end of the 1G era, digital technologies began to emerge as a successor to analog systems. The transition to digital paved the way for more advanced and efficient mobile communication systems.



Evolution of 2G

1. GSM (Global System for Mobile Communications):
 - GSM, introduced in the early 1990s, was a major milestone in the 2G evolution.
 - It shifted from analog to digital communication, using a combination of Time Division Multiple Access (TDMA) and Frequency Division Multiple Access (FDMA) to allow multiple users to share the same frequency band.
1. Digital Voice and Encryption:
 - 2G networks brought digital voice encoding, significantly improving call quality and reducing interference.
 - Enhanced encryption was introduced to improve the security of communication, making it more difficult for unauthorized parties to intercept conversations.
1. SMS (Short Message Service):
 - One of the notable features introduced with 2G was SMS, allowing users to send short text messages between mobile devices.
 - SMS quickly gained popularity and became a ubiquitous form of communication.
1. Data Services:
 - While primarily designed for voice communication, 2G networks started to support data services at low speeds.
 - GPRS (General Packet Radio Service) and EDGE (Enhanced Data rates for GSM Evolution) were introduced as enhancements to GSM, providing limited data capabilities.
1. Global Roaming:
 - 2G networks facilitated global roaming, allowing users to use their mobile devices in different countries with compatible networks.
1. SIM Cards:
 - Subscriber Identity Module (SIM) cards became a standard feature in 2G networks. These removable cards contained subscriber information and allowed users to switch devices easily.
1. Introduction of Multimedia Messaging (MMS):
 - Towards the end of the 2G era, some networks began supporting Multimedia Messaging Service (MMS), enabling the exchange of multimedia content such as pictures and videos.



Evolution of 3G

1. UMTS (Universal Mobile Telecommunications System):
 - UMTS, based on the WCDMA (Wideband Code Division Multiple Access) air interface, was one of the first 3G technologies.
 - UMTS significantly increased data transfer speeds compared to 2G technologies, allowing for enhanced multimedia services.
 2. HSPA (High-Speed Packet Access):
 - HSPA is an enhancement of UMTS, offering higher data rates through the introduction of new protocols.
 - HSDPA (High-Speed Downlink Packet Access) improved downlink speeds, while HSUPA (High-Speed Uplink Packet Access) improved uplink speeds.
 3. HSPA+ (Evolved High-Speed Packet Access):
 - HSPA+ further extended the capabilities of HSPA, providing even higher data rates.
 - It introduced multiple input, multiple output (MIMO) technology and higher order modulation schemes.
 4. EV-DO Rev. B (Evolution-Data Optimized Revision B):
 - While CDMA2000 and EV-DO were primarily associated with 3G in CDMA networks, EV-DO Rev. B was an upgrade that increased data rates and system capacity.
 5. 4G LTE (Long-Term Evolution):
 - LTE is often considered part of the 3G to 4G transition, offering a substantial leap in data rates and network efficiency.
 - LTE provides all-IP (Internet Protocol) based communication, low-latency connectivity, and improved spectral efficiency.
- The introduction of LTE Advanced and LTE Advanced Pro continued to enhance 4G capabilities, offering features like Carrier Aggregation, enhanced MIMO, and improved modulation schemes.



Evolution of 4G

1. LTE (Long-Term Evolution):
 - LTE is the technology that forms the foundation of 4G networks.
 - It introduced significant improvements in data transfer rates, reduced latency, and enhanced spectral efficiency.
 - LTE networks provided a substantial boost in download and upload speeds compared to 3G technologies.
2. LTE-Advanced (LTE-A):
 - LTE-A is an enhancement of the original LTE standard, introducing additional features to improve performance.
 - Carrier Aggregation: Allows the combination of multiple LTE carriers to increase data rates and capacity.
 - MIMO (Multiple Input, Multiple Output): Utilizes multiple antennas for improved data throughput and coverage.
 - Higher Modulation Schemes: Supports more advanced modulation techniques for increased data rates.
3. LTE-Advanced Pro:
 - LTE-Advanced Pro, also known as 4.5G or 4.9G, represents further enhancements beyond LTE-A.
 - Enhanced Carrier Aggregation: Supports more carrier aggregation combinations for even higher data rates.
 - Massive MIMO: Utilizes a large number of antennas to enhance spectral efficiency and network capacity.
 - Improved IoT (Internet of Things) Support: Enhanced support for IoT devices with lower power consumption and extended coverage.
4. Gigabit LTE:
 - Gigabit LTE represents a milestone in achieving peak download speeds of up to 1 Gbps.
 - Utilizes advanced technologies such as 4x4 MIMO, 256-QAM modulation, and carrier aggregation to achieve higher data rates.
5. 5G Introduction:
 - While 5G is considered a distinct generation of mobile technology, its introduction and coexistence with 4G are crucial in the evolution of mobile networks.
 - 5G networks bring significantly higher data rates, lower latency, and the ability to connect a massive number of devices simultaneously.
 - Initial 5G deployments may involve non-standalone (NSA) architectures, where 5G and 4G networks work together to provide enhanced services.
6. Dual Connectivity:
 - Dual Connectivity is a feature that allows devices to simultaneously connect to both 4G and 5G networks, providing improved performance and a smoother transition to full 5G coverage.



Evolution of 5G

- 1. Pre-5G Technologies:**
 - Before the formal introduction of 5G, there were enhancements and technologies in the 4G LTE (Long-Term Evolution) era, such as LTE Advanced (LTE-A) and LTE Advanced Pro, that provided stepping stones toward the capabilities of 5G.
- 2. Release 15 - 5G NR (New Radio):**
 - In December 2017, the 3rd Generation Partnership Project (3GPP) released the first official 5G standard, known as Release 15.
 - 5G NR introduced new radio technologies, frequency bands, and massive MIMO (Multiple-Input, Multiple-Output) for improved spectral efficiency.
 - The initial focus was on enhanced Mobile Broadband (eMBB), providing faster data rates and increased network capacity.
- 3. Release 16 - Expanded Capabilities:**
 - Released in July 2020, 3GPP's Release 16 expanded the 5G capabilities to support a broader range of use cases and industries.
 - It introduced features such as URLLC (Ultra-Reliable Low Latency Communication) for applications with stringent latency requirements, and mMTC (massive Machine Type Communication) for connecting a massive number of IoT devices.
 - Integrated access technologies like Wi-Fi and fixed broadband for seamless connectivity.
- 4. Release 17 - Further Enhancements:**
 - 3GPP's Release 17, expected to be completed in 2022, is anticipated to bring further enhancements to 5G.
 - It aims to address more advanced use cases, improve network efficiency, and continue evolving the 5G ecosystem.
- 5. Deployment and Commercialization:**
 - Commercial deployments of 5G networks began around 2019, with various countries and telecom operators rolling out 5G services in stages.
- 6. Standalone (SA) 5G Networks:**
 - Initially, 5G networks were deployed in a non-standalone (NSA) mode, relying on existing 4G infrastructure for certain functions. However, the move to standalone (SA) 5G networks, with a fully independent architecture, has become a key focus.
- 7. Network Slicing and Edge Computing:**
 - 5G enables network slicing, allowing operators to create virtualized, customized networks to meet specific requirements for different services and industries.
 - Edge computing is integrated into 5G networks, bringing computational resources closer to the end-users, reducing latency and supporting applications like augmented reality and autonomous vehicles.

Prototype: -



